
Learning Generative Models with Sinkhorn Divergences

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Algorithm 2 Sinkhorn loss $\hat{W}_{\varphi, \varepsilon}^{(L)}(x_1^m, y_1^m)$

Input: $\theta, (x_i)_{i=1}^m, (y_j)_{j=1}^m, \varepsilon$

Output: w

$$\forall (i, j), \hat{c}_{i,j} \stackrel{\text{def.}}{=} \|f_{\varphi}(x_i) - f_{\varphi}(y_j)\|$$

$$K_{i,j} = e^{-\frac{\hat{c}_{i,j}}{\varepsilon}}$$

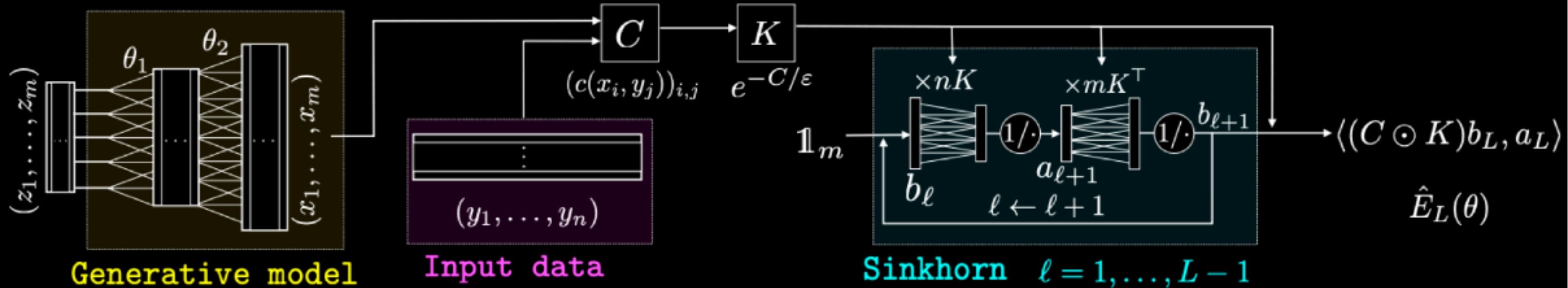
$$b \leftarrow \mathbb{1}_n,$$

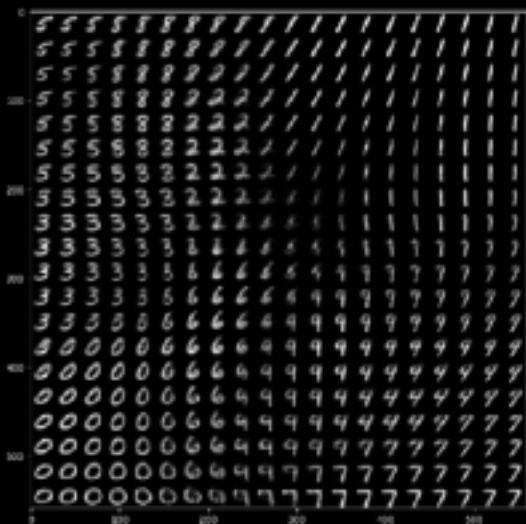
for $\ell = 1, 2, \dots, L$ **do**

$$a \leftarrow \frac{\mathbb{1}_n}{Kb}, b \leftarrow \frac{\mathbb{1}_n}{K^{\top}a}$$

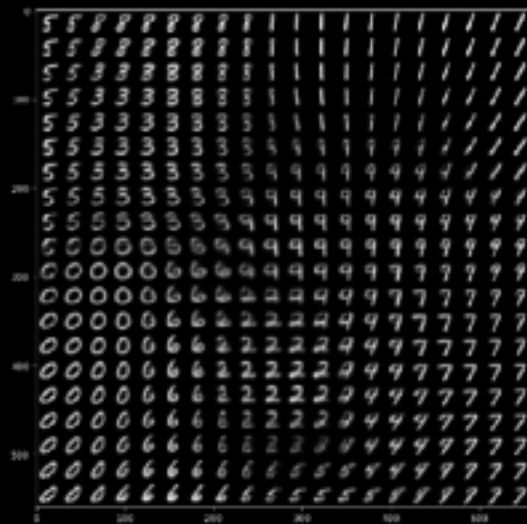
end for

return $w = \langle (K \odot \hat{c})b, a \rangle$ (see (8))

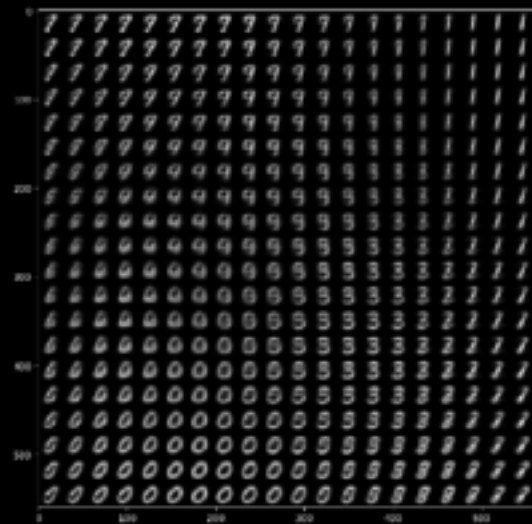




(a) $\varepsilon = 1$
 $m = 200, L = 10$



(b) $\varepsilon = 10^{-1}$
 $m = 200, L = 100$



(c) $\varepsilon = 10^{-1}$
 $m = 10, L = 300$

no need to train adversary/witness

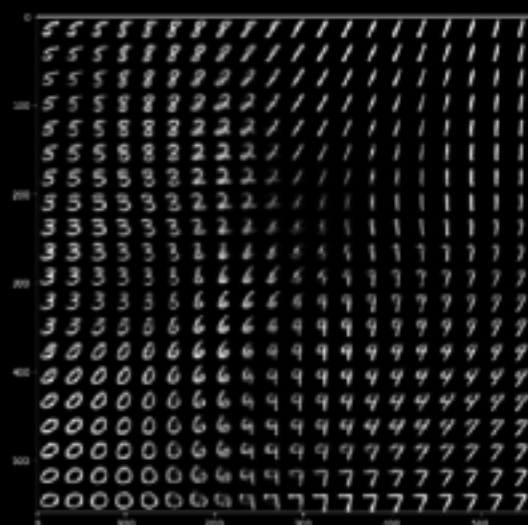
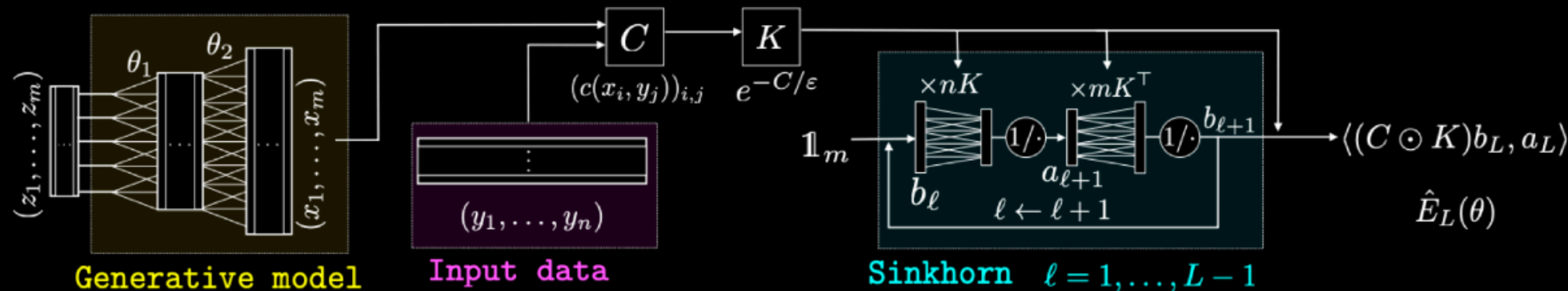
Application: Generative Modeling

Learning Generative Models with Sinkhorn Divergences

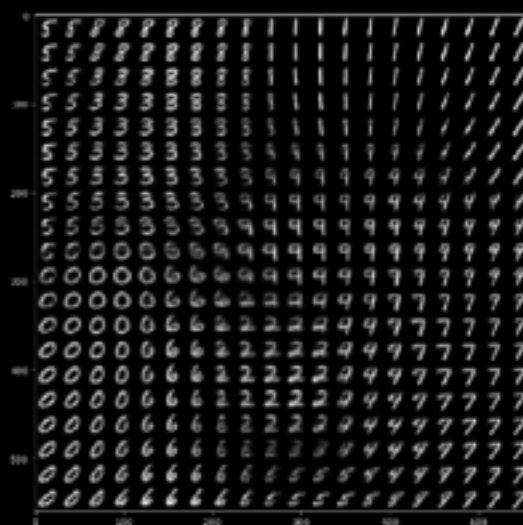
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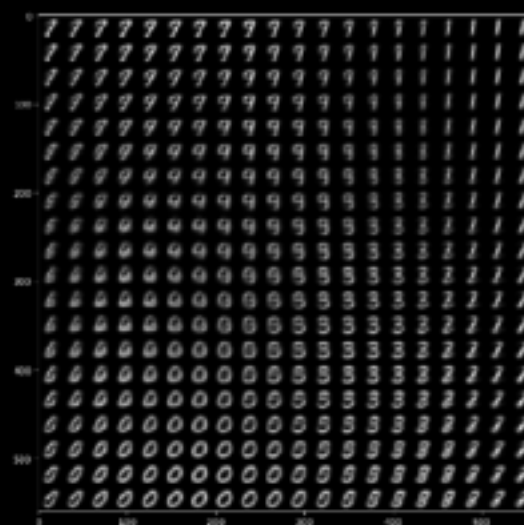
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(a) $\varepsilon = 1$
 $m = 200, L = 10$



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 $m = 10, L = 300$

Algorithm 1 SGD with Auto-diff

Input: $\theta_0, \varphi_0, (y_j)_{j=1}^n$ (the real data), m (batch size), L (number of Sinkhorn iterations), ε (regularization parameter), α (learning rate)

Output: θ, φ

$\theta \leftarrow \theta_0, \varphi \leftarrow \varphi_0,$

for $k = 1, 2, \dots$ **do**

for $t = 1, 2, \dots, n_c$ **do**

 Sample $(y_j)_{j=1}^m$ from the observations.

 Sample $(z_i)_{i=1}^m \stackrel{\text{i.i.d.}}{\sim} \zeta, (x_i)_{i=1}^m \stackrel{\text{def.}}{=} g_\theta(z_1^m)$

$\text{grad}_\varphi \leftarrow \text{AutoDiff}_\varphi \left(2\hat{W}_{\varphi, \varepsilon}^{(L)}(x_1^m, y_1^m) - \hat{W}_{\varphi, \varepsilon}^{(L)}(x_1^m, x_1^m) - \hat{W}_{\varphi, \varepsilon}^{(L)}(y_1^m, y_1^m) \right)$

$\varphi \leftarrow \varphi + \alpha \text{RMSProp}(\text{grad}_\varphi).$

$\varphi \leftarrow \text{clip}(\varphi, -c, c)$

end for

 Sample $(y_j)_{j=1}^m$ from the observations.

 Sample $(z_i)_{i=1}^m \stackrel{\text{i.i.d.}}{\sim} \zeta, (x_i)_{i=1}^m \stackrel{\text{def.}}{=} g_\theta(z_1^m)$

$\text{grad}_\theta \leftarrow \text{AutoDiff}_\theta \left(2\hat{W}_{\varphi, \varepsilon}^{(L)}(x_1^m, y_1^m) - \hat{W}_{\varphi, \varepsilon}^{(L)}(x_1^m, x_1^m) - \hat{W}_{\varphi, \varepsilon}^{(L)}(y_1^m, y_1^m) \right)$

$\theta \leftarrow \theta - \alpha \text{RMSProp}(\text{grad}_\theta).$

end for

Advantage over GAN/WGAN: **no need to train adversary/witness** — just generator + loss layer!

Application: Knowledge Distillation

Sinkhorn Distance Minimization for Knowledge Distillation

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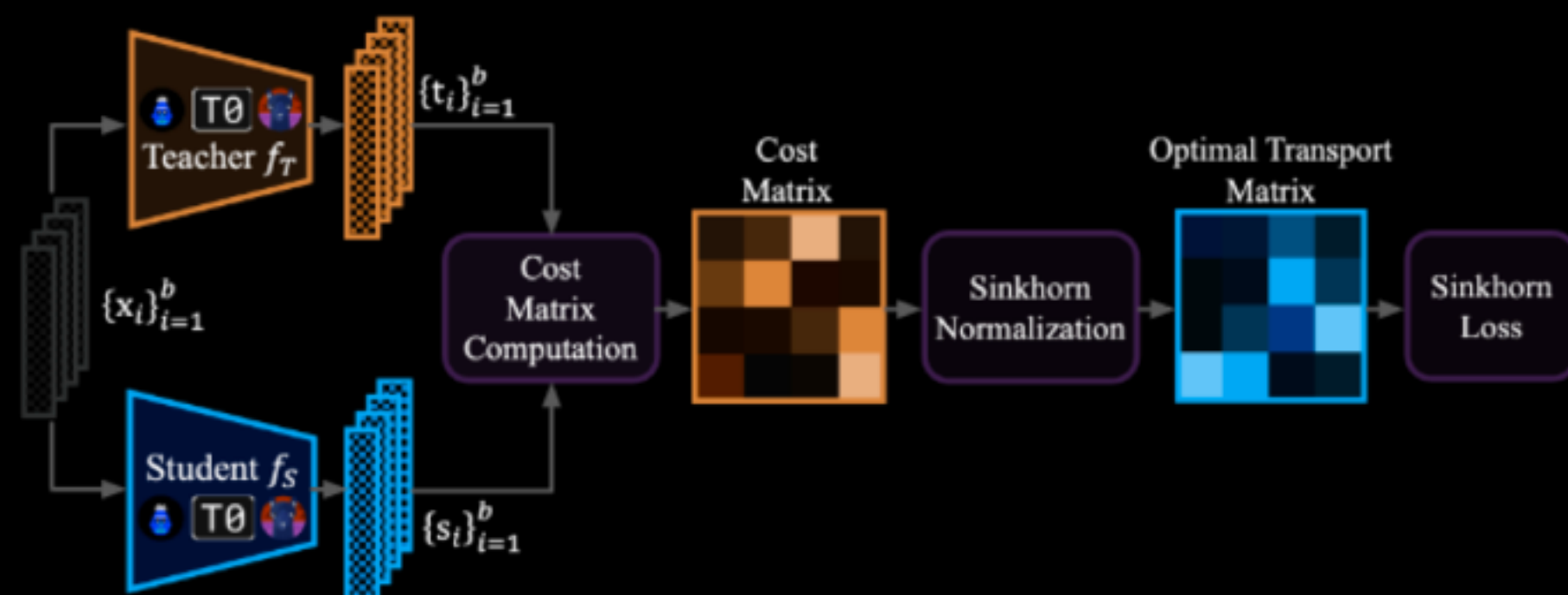
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Abstract

Knowledge distillation (KD) has been widely adopted to compress large language models (LLMs). Existing KD methods investigate various divergence measures including the Kullback-Leibler (KL), reverse Kullback-Leibler



3.1. Problem Statement

Given a sample $\mathbf{x}_i$ and its ground-truth label $\mathbf{y}_i \in \mathbb{R}^d$ in the training set, the output logits with softmax activation σ_τ from the teacher f_T and the student f_S are respectively $\mathbf{t}_i \in \mathbb{R}^d$ and $\mathbf{s}_i \in \mathbb{R}^d$:

$$\mathbf{t}_i = \sigma_\tau(f_T(\mathbf{x}_i)), \quad \mathbf{s}_i = \sigma_\tau(f_S(\mathbf{x}_i)), \quad (1)$$

where τ is the temperature and d is the dimension of the output logits. The objective of KD is to minimize the measured divergence $J(\mathbf{t}_i, \mathbf{s}_i)$.

Method	#Params.	COLA (MCC)	SST-2 (ACC)	MNLI-(m/mm) (ACC)	MRPC (F1)	RTE (ACC)	QNLI (ACC)	QQP (ACC)	Avg
BERT _{base} (T) (Devlin et al., 2018)	110M	60.3	93.7	84.9/84.8	91.4	71.1	91.7	91.5	83.28
BERT _s (S) (Turc et al., 2019)	66M	51.2	91.0	81.7/82.6	89.2	66.1	89.3	90.4	79.53
Task-specific Representation-based Distillation									
PKD (Sun et al., 2019)	66M	45.5	91.3	81.3/-	85.7	66.5	88.4	88.4	77.63
TinyBERT (Jiao et al., 2020)	66M	53.8	92.3	83.1/83.4	88.8	66.5	89.9	90.5	80.3
RKD (Park et al., 2019)	66M	53.4	91.7	-	86.1	68.6	89.5	90.9	80.03
CKD (Park et al., 2021b)	66M	55.1	93.0	83.6/84.1	89.6	67.3	90.5	91.2	81.11
SFTN (Park et al., 2021a)	66M	53.6	91.5	-	85.3	68.5	89.5	90.4	79.80
TAKD (Mirzadeh et al., 2020)	66M	53.8	91.4	-	85.0	68.5	89.6	90.7	79.83
ProKT (Shi et al., 2020)	66M	54.3	91.3	-	86.3	68.4	89.7	90.9	80.15
MGSKD (Liu et al., 2022)	66M	49.1	91.7	83.3/83.9	89.8	67.9	90.3	91.2	80.00
MetaDistill (Zhou et al., 2022)	66M	58.6	92.3	-	86.8	69.4	90.4	91.0	81.42
AD-KD (Wu et al., 2023)	66M	58.3	91.9	83.4/84.2	91.2	70.9	91.2	91.2	82.45
Task-specific Logits-based Distillation									
Vanilla KD (Hinton et al., 2015)	66M	53.6	91.1	82.7/83.1	89.4	66.8	90.1	90.5	80.25
RCO (Jin et al., 2019)	66M	53.6	91.4	-	85.1	67.6	89.7	90.6	79.67
DML (Zhang et al., 2018)	66M	53.7	91.5	-	85.1	68.4	89.6	90.3	79.77
PD (Turc et al., 2019)	66M	-	91.1	82.5/83.4	89.4	66.7	89.4	90.7	-
ReAugKD (Zhang et al., 2023)	66M	59.4	92.5	-	86.3	70.4	90.7	91.2	81.75
SinKD (ours)	66M	60.2	93.1	83.8/84.2	91.3	71.1	90.5	91.3	82.92